The Geometrical Origin of Quantum Mechanics: Planck Scale, Dirac Equation, and Superconductivity

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Abstract

Within the geometrical framework of the Unified Field Theory, this paper further extends the results of Paper D and systematically reveals the deeper geometrical origin of quantum mechanics.

Starting from the principle that the reduced Planck constant $\hbar$ represents the action per radian, we derive a complete set of Planck natural units (length, time, mass, energy, frequency, density, etc.) and demonstrate that all of them are determined by a closed loop formed by the three fundamental constants $\{c, \mu_g, \hbar\}$.

On this basis, by introducing Lorentz covariance and the linearization principle, the Dirac equation

$$\left(i\hbar\gamma^{\mu}\partial_{\mu} - mc\right)\Psi = 0$$

is obtained, revealing the inevitable correspondence between spin-1/2 and geometrical 4π periodicity.

Furthermore, by employing geometrical scaling and Fourier duality, we rigorously derive the Heisenberg uncertainty principle

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}$$

showing that it is not an empirical postulate but a direct consequence of the action-per-radian scaling.

The paper also provides a unified geometrical explanation of quantum phenomena such as angular momentum quantization, discrete energy levels, geometrical phase, and spin periodicity.

Finally, extending the geometrical framework to macroscopic quantum systems, we present a geometrical interpretation of superconductivity, including phase coherence locking, flux quantization, and the vanishing of electrical resistance.

The study establishes a closed logical chain:

Geometrical scaling → Planck units → Dirac equation → Uncertainty principle → Macroscopic quantum phenomena,

thereby unveiling the geometrical foundation of the entire structure of quantum mechanics.

Part I: Geometrical Derivation of the Planck System

1.1 Preliminaries: The Minimal Set of Three Constants

Within the geometrical framework of the Unified Field Theory, all Planck units no longer depend on the traditional set$c, G, \hbar$. Instead, they are determined by a closed loop formed by the three constants:

$$\{c,\ \mu_g,\ \hbar\}$$

Through the geometry–gravity bridging relation:

$$\text{(E3)} \qquad G = \frac{4\pi c^2}{\mu_g}$$

the gravitational constant G is no longer an independent parameter, but is determined jointly by μ_g and c.

1.2 Planck Length and Planck Time

Traditional definition:

$$l_P = \sqrt{\frac{\hbar G}{c^3}},$$

$$t_P = \frac{l_P}{c}.$$

Geometric expression: substituting (E3):

$$l_P = \frac{4\pi\hbar}{\mu_g c},$$

$$t_P = \frac{4\pi\hbar}{\mu_g c^3}.$$

Dimensional check:

- $[\mu_g] = \mathrm{kg/m}$, $[\hbar] = \mathrm{kg{\cdot}m^2/s}$, $[c] = \mathrm{m/s}$.
- $\hbar/(\mu_g c) \sim (\mathrm{kg{\cdot}m^2/s})/(\mathrm{kg/m \cdot m/s}) = \mathrm{m^2}$.
- After taking the square root, the dimension is meters, consistent with l_P.
- $t_P = l_P/c$ has dimension seconds.

1.3 Planck Mass and Geometric Mass

Traditional definition:

$$M_P = \sqrt{\frac{\hbar c}{G}}$$

Geometric expression:

$$M_P = \frac{1}{2\pi}\sqrt{\mu_g \hbar c}$$

Further, define the Geometric Planck Mass:

$$M_{geo} = \mu_g l_P = 4\pi M_P,$$

showing that under geometrical normalization, a natural 4π factor appears.

Dimensional check:

$[\mu_g l_P] = (\mathrm{kg/m}) \cdot \mathrm{m} = \mathrm{kg}$, consistent with mass.

1.4 Planck Energy and Frequency

Energy:

$$E_P = M_P c^2 = \frac{1}{2\pi}\sqrt{\mu_g \hbar c^3}$$

Frequency (traditional $\omega_P = \sqrt{c^5/\hbar G}$):

Substituting $G = 4\pi c^2/\mu_g$:

$$\omega_P = \frac{\mu_g c^3}{4\pi\hbar}$$

Within the geometric closed loop, there exists the bridging relation:

$$\Omega = \frac{\omega_P}{4\pi}$$

i.e., the geometric rotational scale Ω differs from the Planck frequency by only a 4π normalization.

1.5 Planck Density and Energy Density

Planck density:

$$\rho_P = \frac{M_{geo}}{l_P^3} = \frac{\mu_g^2 c}{4\pi\hbar}$$

Energy density:

$$\varepsilon_P = \rho_P c^2 = \frac{\mu_g^2 c^3}{4\pi\hbar}$$

1.6 Summary

- All Planck quantities (length, time, mass, energy, frequency, density) can be derived from the closed set of three constants $\{c, \mu_g, \hbar\}$.
- The factor 4π serves as a geometric normalization ensuring dimensional and topological consistency.
- Particularly significant results are:

$$M_{geo} = 4\pi M_P,$$

$$M_* = \sqrt{\frac{\mu_g \hbar}{c}} = \sqrt{4\pi} M_P,$$

which demonstrate that the Planck system acquires a natural normalization correction under the geometrical framework of the Unified Field Theory.

Part II: Dirac Equation and the Geometrical Interpretation of Spin

2.1 Relativistic Energy–Momentum Relation

In special relativity, a particle satisfies the energy–momentum relation:

$$(E13) \quad E^2 = p^2c^2 + m^2c^4$$

Dimensional check:

$[E] = \mathrm{J}, [p] = \mathrm{kg{\cdot}m/s}, [m] = \mathrm{kg}$. The relation is dimensionally consistent.

2.2 Klein–Gordon Equation (Second-Order Form)

With the replacements E $\to i\hbar\,\partial_t, p \to -i\hbar$, the scalar wave equation is obtained:

$$(E14) \quad \left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2 + \frac{m^2c^2}{\hbar^2}\right)\phi = 0$$

However, this equation is of second order, and its probability density is not positive-definite.

2.3 Dirac's Linearization Principle

Dirac required a first-order equation, whose square would return to (E13). Assume:

$$(E15) \quad i\hbar\,\partial_t\,\Psi = \backslash Big\big(c\,\alpha\cdot p + \beta mc^2\backslash Big\big)\,\Psi,$$

where:

- Ψ: multi-component wave function,
- α, β: matrices to be determined,
- $p = -i\hbar\nabla$.

For $(E15)^2$ to recover (E13), the matrices must satisfy:

$$(E16) \quad \{\alpha_i, \alpha_j\} = 2\delta_{ij}, \quad \{\alpha_i, \beta\} = 0, \quad \beta^2 = 1.$$

This is the Clifford algebra relation. The minimal irreducible representation is 4×4,

meaning that Ψ must be a four-component spinor.

2.4 Covariant Form

Define:

$$\gamma^0 = \beta, \quad \gamma^i = \beta\alpha_i \quad (i = 1,2,3),$$

satisfying:

$$(E17) \quad \{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu},$$

$$\eta^{\mu\nu} = \mathrm{diag}(1, -1, -1, -1).$$

Thus the Dirac Equation is obtained:

$$(E18) \quad \left(i\hbar\gamma^\mu \partial_\mu - mc\right)\Psi = 0.$$

2.5 Probability Current and Conservation

Define the Dirac adjoint $\bar{\Psi} = \Psi^\dagger\gamma^0$. Then the probability current is:

$$(E19) \quad j^\mu = c\bar{\Psi}\gamma^\mu\Psi, \qquad \partial_\mu j^\mu = 0$$

Here $j^0 = \Psi^\dagger\Psi \geq 0$, which is a positive-definite probability density.

2.6 Spin and Geometrical 4π Periodicity

Within the geometrical scale framework:

- For integer spin, the single-valued condition is $\psi(\theta + 2\pi) = \psi(\theta)$.
- For half-integer spin, the geometry allows $\psi(\theta + 2\pi) = -\psi(\theta)$, so the state returns to itself only after a 4π rotation.

This is the manifestation of the fact that SU(2) double-covers SO(3).

Therefore:

Spin 1/2 is not an empirical setting, but the inevitable result of $\hbar$ =scale of action per radian + geometric duality.

2.7 Non-Relativistic Limit

In the low-velocity approximation:

- The Dirac equation → Pauli equation, introducing the spin–magnetic moment term $\sigma \cdot B$.
- In an even lower-energy limit → Schrödinger equation (E5).

Thus, Schrödinger and Dirac equations are not independent, but expansions of the same geometrical framework at different energy scales.

2.8 Summary

- The Dirac equation (E18) naturally emerges from the geometrical framework of the Unified Field Theory.
- The essence of spin 1/2 is the geometrical 4π periodicity, no longer a manually imposed quantum number.
- The geometrical scale $\hbar$ and the speed of light c permeate the equation's structure, while the mass term can be expressed in terms of the geometric mass (M_*), eliminating the need for explicit introduction of G.

Part III: Geometrical Origin of the Heisenberg Uncertainty Principle

3.1 Geometrical Scale and Conjugate Variables

Within the geometrical framework of the Unified Field Theory:

- $\hbar$ represents the minimal action scale per radian.
- For a conjugate pair (x, p), the phase can be written as:

$$\varphi = \frac{1}{\hbar}\int p\ dx,$$
$$\psi(x) = A(x)e^{i\varphi(x)}.$$

Thus, $\hbar$ sets the minimal unit of action between position and momentum as a conjugate pair.

3.2 Commutation Relation

From the scale defined by $\hbar$, the fundamental commutation relation between position and momentum is obtained:

$$(E20) \quad [x, p] = i\hbar$$

Dimensional check:

$[x] = \mathrm{m}, [p] = \mathrm{kg{\cdot}m/s},\ so\ the\ RHS\ [i\hbar] = kg{\cdot}m^2/s,$ consistent with the LHS.

3.3 Robertson–Schrödinger Inequality

For any state $|\psi\rangle$, the general inequality holds:

$$\Delta A^2\,\Delta B^2 \geq \left(\frac{1}{2i}\langle[\hat{A}, \hat{B}]\rangle\right)^2 + Cov(A, B)^2$$

Taking $\hat{A} = \hat{x}, \hat{B} = \hat{p}$, and using (E20):

$$(E21) \quad \Delta x \, \Delta p \geq \frac{\hbar}{2}$$

3.4 Fourier-Scale Derivation (Intuitive Approach)

In position representation:

$$\psi(x) = \frac{1}{\sqrt{2\pi}} \int \tilde{\psi}\,(k)\, e^{ikx}\, dk, \quad p = \hbar k.$$

Fourier analysis gives:

$$\Delta x \, \Delta k \;\geq\; \frac{1}{2}$$

Multiplying both sides by $\hbar$:

$$(E22) \quad \Delta x \, \Delta p \geq \frac{\hbar}{2}$$

This shows that the uncertainty relation arises directly from the action scale $\hbar$ and the Fourier duality structure.

3.5 Saturation Condition and Minimum-Uncertainty State

For a Gaussian wave packet:

$$\psi(x) = \frac{1}{(2\pi\sigma_x^2)^{1/4}} \exp\left(-\frac{(x - x_0)^2}{4\sigma_x^2} + \frac{i}{\hbar} p_0 (x - x_0) \right)$$

one finds:

$$\Delta x = \sigma_x, \quad \Delta p = \frac{\hbar}{2\sigma_x},$$

Thus:

$$(E23) \quad \Delta x \, \Delta p = \frac{\hbar}{2}$$

This means the Gaussian wave packet is a minimum-uncertainty state, geometrically corresponding to the simplest "closed loop" scale.

3.6 Conceptual Interpretation

- Traditional view: The uncertainty principle is a limitation of measurement.
- Geometrical view: Uncertainty arises because $\hbar$ is the fundamental scale of action per radian:
- The more precise the position, the more delocalized the momentum phase.
- The more precise the momentum, the more delocalized the position coordinate.

Thus, the uncertainty relations (E21, E22) are not externally imposed postulates, but inevitable consequences of geometrical–topological scaling.

3.7 Summary

- From both the commutation relation (E20) and Fourier scaling, two independent derivations lead to (E21) and (E22).
- $\hbar$, as the "action per radian" scale, explains why $\Delta x \quad \Delta p$ cannot be smaller than $\hbar/2$.
- Therefore, the Heisenberg uncertainty principle obtains a geometrical foundation rather than being an empirical rule.

Part IV: Other Phenomena in Quantum Mechanics

4.1 Quantization of Angular Momentum

The condition of single-valuedness requires that the action along a closed loop equals an integer multiple of $2\pi\hbar$:

$$\oint \nabla\varphi \cdot dl = 2\pi\hbar$$

Thus angular momentum takes discrete values:

$$(E24) \quad L_z = m\hbar, \quad m \in Z.$$

4.2 Discreteness of Energy Levels

In the harmonic oscillator system, the solution gives:

$$(E25) \quad E_n = (n + \backslash tfrac12)\hbar\omega,$$
$$n = 0,1,2, \ldots$$

Here the quantization of energy arises directly from the scale of $\hbar$.

For the hydrogen atom, the energy spectrum is:

$$(E26) \quad E_n = -\frac{me^4}{2(4\pi\epsilon_0)^2\hbar^2 n^2}$$

This also shows that the spacing of energy levels is proportional to $\hbar^{-2}$, highlighting its role as a fundamental scale.

4.3 Geometrical Phase

In the presence of a gauge potential A, the accumulated phase of a wave function is:

$$(E27) \quad \Delta\varphi = \frac{1}{\hbar}\oint \mathrm{b}A \cdot d\mathrm{l}$$

Examples:

- Aharonov–Bohm effect: Electrons acquire a phase shift even in a field-free region due to the vector potential.
- Berry phase: A closed loop in parameter space leads to an additional geometrical phase.

These effects illustrate that $\hbar$ is the fundamental scale of geometrical–topological effects.

4.4 The 4π Periodicity of Spin

The geometrical scaling of the Unified Field Theory permits two kinds of single-valuedness:

- For integer spin: $\psi(\theta + 2\pi) = \psi(\theta)$.
- For half-integer spin: $\psi(\theta + 2\pi) = -\psi(\theta)$, and only after 4π does the wave function return to its original state.

Thus:

$$(E28) \quad \text{Spin-}1/2 \leftrightarrow SU(2) \text{ double cover of } SO(3).$$

The 4π periodicity of spin is therefore not an empirical assumption but a necessary consequence of spatial geometrical double-valuedness.

4.5 Wave–Particle Duality and Interference Phenomena

The de Broglie relations:

$$p = \hbar k, \quad E = \hbar\omega,$$

show that particle trajectories correspond to geometrical projections of wave phases.

In the double-slit experiment, the spacing of interference fringes is given by:

$$\Delta y = \frac{\lambda L}{d}, \quad \lambda = \frac{h}{p},$$

directly reflecting the role of $\hbar$ as the fundamental scale.

4.6 Summary

- Angular momentum quantization (E24), discrete energy levels (E25, E26), geometrical phases (E27), spin periodicity (E28), and wave–particle duality are all unified under the interpretation of $\hbar$ as "action per radian."
- What traditional quantum mechanics regarded as empirical rules are here revealed, under the geometrical framework of the Unified Field Theory, to be inevitable consequences.

Part V: Geometrical Explanation of Superconductivity

5.1 Macroscopic Wavefunction and Coherence

In a superconducting system, a large number of electrons form Cooper pairs. Their coherence can be described by a macroscopic wavefunction:

$$\Psi(\mathbf{r}, t) = \sqrt{\rho(\mathbf{r}, t)}\, e^{iS(\mathbf{r},t)/\hbar}$$

Here, once again, $\hbar$ functions as the "action per radian," ensuring that macroscopic phases can accumulate and lock coherently.

5.2 Josephson Effect

In a superconducting tunneling junction, the phase difference $\Delta\varphi$ between two superconductors leads to a current:

$$(E29) \quad I = I_c \sin\left(\frac{\Delta\varphi}{\hbar}\right)$$

This shows that the current is driven by the phase difference of action, reflecting the role of geometrical single-valuedness in macroscopic quantum systems.

5.3 Magnetic Flux Quantization

In a superconducting loop, the requirement of single-valuedness gives:

$$\oint \nabla\varphi \cdot d\mathrm{l} = 2\pi\hbar$$

Combining this with the gauge phase of charge $2e$, one obtains magnetic flux quantization:

$$(E30) \quad \Phi = n\frac{h}{2e}, \quad n \in Z.$$

Here, $h = 2\pi\hbar$ corresponds precisely to the closed-loop action of a full cycle.

5.4 Geometrical Explanation of Zero Resistance

- In normal conductors, electron scattering destroys phase coherence.
- In superconductors, geometrical constraints enforce global phase consistency:
- Current corresponds to the phase gradient.
- Phase rigidity guarantees dissipationless transport.
- Thus, the macroscopic manifestation is zero resistance.

5.5 Critical Conditions and Geometrical Predictions

- Critical current: When the phase gradient induced by an external field exceeds the geometrical scale ($\Delta\varphi/\hbar \sim 1$), coherence is destroyed.
- Critical magnetic field: If the external magnetic flux fails to satisfy (E30), the phase cannot close, leading to the disappearance of superconductivity.
- Critical temperature: Competition arises between the geometrical scale and thermal energy; when thermal excitations $k_B T$ exceed the pairing energy, wavefunction coherence collapses.

5.6 Summary

- Superconducting phenomena — Josephson effect, current–phase relation, magnetic flux quantization, and zero resistance — all follow naturally from the geometrical scale of $\hbar$ and the condition of phase single-valuedness.
- This indicates that macroscopic quantum effects are not independent empirical rules, but rather extensions of the geometrical necessity of the Unified Field Theory.
- Superconductivity can thus be understood as: the geometrical scale enforcing macroscopic phase coherence, thereby eliminating scattering and dissipation.

Part VI: Conclusion and Outlook

6.1 Review of the Logical Closed Loop

Within the geometrical framework of the Unified Field Theory, starting from the essence of $\hbar =$ action per radian, we gradually uncover the deeper unified logic of quantum mechanics:

1. Planck System: All Planck units $(\mathrm{l}_P, t_P, M_P, E_P, \omega_P, \rho_P, etc.)$ are derived from the closed $set\ \{c, \mu_g, \hbar\}$.

2. Dirac Equation: By requiring Lorentz covariance and linearization, the Dirac equation emerges naturally from the relativistic energy–momentum relation. The 4π periodicity of spin-1/2 becomes a geometrical necessity.

3. Heisenberg Uncertainty Principle: From both commutation relations and Fourier scaling, the inequality $\Delta x \ \Delta p \geq \hbar/2$ is obtained, highlighting the scale-setting role of $\hbar$.

4. Quantum Mechanical Phenomena: Angular momentum quantization, discrete energy levels, geometrical phase, spin double covering, and wave–particle duality are all inevitable consequences of single-valuedness and action scaling.

5. Superconductivity: Macroscopic coherent wavefunctions lead to vanishing resistance, flux quantization, and the Josephson effect — a geometrical interpretation of macroscopic quantum phenomena.

6.2 Conceptual Elevation

- Action Scale: $\hbar$ is the minimal unit of action in the physical world. Its geometrical nature guarantees the universality of quantum phenomena.
- Geometry and Topology Determine Quantum Laws: All quantum effects can be traced back to geometrical closed loops, phase single-valuedness, and 4π periodicity.
- Unification: Quantum mechanics is no longer a collection of empirical postulates, but the necessary consequence of spatial geometry, mass coupling, and the action scale.

6.3 Directions for Experimental Verification

1. Reinterpretation of Planck Units: High-precision measurements of c and $\hbar$ can test the closed loop of the three constants and the geometrical normalization factor.

2. Spin and Geometrical Periodicity: Experimental verification of the topological explanation of electron spin 4π periodicity.

3. Superconductivity and Macroscopic Quantum Phenomena: Study of critical temperature, flux quantization, and their dependence on the geometrical action scale.

6.4 Outlook

- Geometrization of Quantum Field Theory: Rewriting gauge fields and interactions within the unified geometrical framework.
- Cosmological Scale: The geometrical reinterpretation of Planck quantities may provide new insights into dark matter and dark energy.
- Technological Applications: If the spatial geometrical scale can be artificially manipulated, it could bring revolutionary breakthroughs in energy, quantum computing, and materials science.

6.5 Final Summary

This work establishes a self-consistent closed loop:

$$\{c, \mu_g, \hbar\} \longrightarrow \{\text{Planck System}\}$$
$$\longrightarrow \text{Dirac Equation}$$
$$\longrightarrow \text{Uncertainty Principle}$$
$$\longrightarrow \text{Quantum Phenomena and Superconductivity.}$$

Conclusion: The foundation of the entire quantum mechanical system lies in geometrical scaling and phase single-valuedness.

This provides a unified explanation for natural constants, the properties of fundamental particles, and macroscopic quantum effects, laying a solid basis for exploring deeper physical laws in the future.

Appendix

Appendix A: Table of Symbols

Symbol	Meaning	Dimension
c	Speed of light	m/s
μ_g	Space–mass coupling constant	kg/m
l_Ω	Rotational scaling length	m
Q	Helical divergence intensity	m^3/s
M	Coupled mass	kg
G	Gravitational constant (derived from μ_g, c)	$m^3/(kg \cdot s^2)$
$\hbar$	Reduced Planck constant (action per radian)	J·s
h	Planck constant (action per full cycle)	J·s
M_P	Planck mass (traditional definition)	kg
M_{geo}	Geometric Planck mass $=4\pi M_P$	kg
M_*	Square-root mass scale = $\sqrt{\mu_g \hbar / c}$	kg
l_P, t_P	Planck length and time	m, s
ω_P	Planck frequency	1/s
Ω	Geometrical rotational scale	1/s
Ψ	Dirac spinor wavefunction	—
γ^μ	Dirac matrices	—
$\rho(r,t)$	Probability density	$1/m^3$
$\varphi(r,t)$	Action phase	J·s
$\psi(r,t)$	Wavefunction	—
V	Potential energy	J
Φ	Magnetic flux	Wb
I	Supercurrent	A

Appendix B: Summary of Formulae and Dimensions

Planck System:

$$\ell_P = \sqrt{\frac{4\pi\hbar}{\mu_g c}},$$

$$t_P = \sqrt{\frac{4\pi\hbar}{\mu_g c^3}}$$

$$M_P = \frac{1}{2\sqrt{\pi}}\sqrt{\frac{\mu_g \hbar}{c}},$$

$$M_{geo} = 4\pi M_P,$$

$$M_* = \sqrt{\frac{\mu_g \hbar}{c}},$$

$$\omega_P = \frac{\mu_g c^3}{4\pi\hbar},$$

$$\Omega = \frac{\omega_P}{4\pi}.$$

Dirac Equation:

$$\left(i\hbar\gamma^\mu \partial_\mu - mc\right)\Psi = 0$$

Uncertainty Principle:

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}$$

Superconductivity:

$$I = I_c \sin\left(\frac{\Delta S}{\hbar}\right),$$
$$\Phi = n\frac{h}{2e}$$

Appendix C: Derivational Details

1. Saturation State of the Uncertainty Principle

Gaussian wave packets satisfy $\Delta x \ \Delta p \ = \ \hbar/2$.

2. Geometrical Interpretation of Spin

$\psi(\theta + 2\pi) = -\psi(\theta)$, $\psi(\theta + 4\pi) = \psi(\theta)$, corresponding to the fact that SU(2) double covers SO(3).

3. Flux Quantization in Superconductors

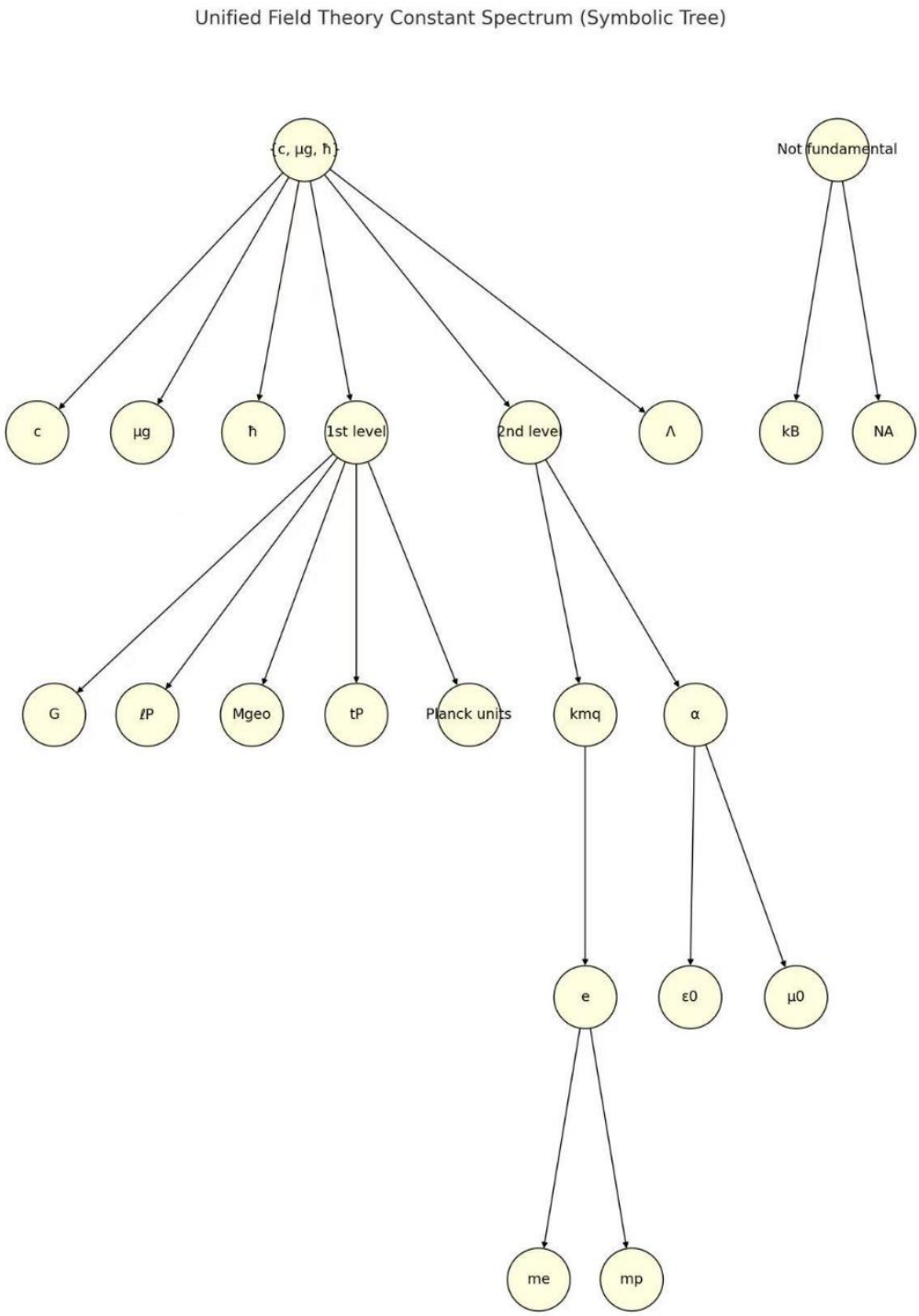
From phase single-valuedness: $\oint \nabla\varphi \cdot dl = 2\pi\hbar$, together with charge $2e, yields$ $\Phi = n\,h\,/\,2e$.

Appendix D: Correspondence with Paper D

- Paper D: Geometrical origin → $\hbar$ → Schrödinger equation.
- Paper F: Extends further → Planck system → Dirac equation → Uncertainty principle → Macroscopic quantum phenomena (superconductivity).
- Together: The two papers form a complete theoretical chain:

$$\{\text{Geometrical scaling, } c, \mu_g, \hbar\}$$
$$\longrightarrow \{\text{Quantum dynamics and Planck limits}\}.$$

Appendix E: Constant Spectrum of the Unified Field Theory



Constant Spectrum Legend

Symbol English meaning

c Speed of light

μg Space – mass coupling constant

ħ Reduced Planck constant

G Gravitational constant

ℓP Planck length

$Mgeo$ Geometric Planck mass

tP Planck time

Planck units Planck units (E, f, T⋯)

kmq Mass – charge conversion constant

α Fine-structure constant

$\varepsilon 0$ Vacuum permittivity

$\mu 0$ Vacuum permeability

e Elementary charge

me Electron mass (state parameter)

mp Proton mass (state parameter)

Λ Cosmological constant (boundary-derived)

kB Boltzmann constant

NA Avogadro constant

Top Level: Three Fundamental Constants

- Speed of light c — space–time standard
- Space–mass coupling constant μ_g— mass standard
- Reduced Planck constant $\hbar$ — quantum action per radian

First-Level Derivatives (directly from the triad)

1. Gravitational constant G

$$G = \frac{4\pi \square^2}{\mu_g}$$

2. Planck length l_P

$$l_P = \sqrt{\frac{4\pi\hbar}{c\mu_g}}$$

3. Geometric Planck mass M_{geo}

$$M_{geo} = \mu_g l_P = 4\pi M_P$$

4. Planck time t_P

$$t_P = \frac{l_P}{c}$$

5. Planck energy, frequency, temperature, etc.

Derived systematically from l_P, M_{geo}, t_P.

Second-Level Derivatives (with electromagnetic bridge)

- Mass–charge conversion constant k_{mq}

Closed with the fine-structure constant α, determines the unit charge.

- Fine-structure constant α

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c}$$

In the unified field theory, uniquely determined from $\{c, \mu_g, \hbar\}$ with the bridge relations.

- Vacuum permittivity ε_0, permeability μ_0

Not independent constants; reconstructed from $\alpha, c, \hbar$.

- Elementary charge e

A coherence parameter of specific states, linked to mass flow through k_{mq}.

Third Level (state-dependent quantities)

- Electron mass m_e, proton mass m_p: state parameters, not fundamental constants.

- Electron charge e: state-dependent, normalized through α and k_{mq}.

Non-fundamental constants (unit conventions)

- Boltzmann constant k_B: conversion between temperature and energy, purely unit definition.
- Avogadro constant N_A: counting unit for chemistry, not a natural constant.

Cosmological constant Λ

- If it exists, it should be determined by the large-scale boundary conditions of the helical space flow, still reducible to the fundamental triad.

Summary

- Minimal triad $\{c, \mu_g, \hbar\}\rightarrow$ generates all physical constants without circular definitions.
- Electromagnetic, gravitational, and quantum constants occupy logical branches of the spectrum.
- Thermodynamic and chemical constants are unit conventions, not part of the fundamental set.

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